NANYANG JUNIOR COLLEGE
Year 2 Preliminary Examination

H2 GEOGRAPHY 9173/02

Paper 2 23 Sept 2025

3 hours

Additional Materials: Insert

READ THESE INSTRUCTIONS FIRST

If you have been given an Answer Booklet, follow the instructions on the front cover of the Booklet.
Write your Centre number, index number and name on the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.
DO **NOT** WRITE ON ANY BARCODES.

Answer Question 1 in **Section A.**
Answer Question 2 in **Section B**.
Answer **one** question in **Section C**.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

Answer Questions 1 and 2 and
EITHER Question 3 **OR** Question 4

Section A

Cluster 4 Fieldwork

1 A group of 10 geography students from a junior college in Singapore undertook an investigation on flood risk in the Klang district of Selangor, Malaysia, during the June holidays.

The students focused on three flood-prone areas: Jalan Tepi Sungei (Site 1), Jalan Tengku Kelana (Site 2) and Kampung Datuk Dagang (Site 3). All 3 sites are located next to the Klang River. They came up with this hypothesis:

Kampung Datuk Degang faces the highest flood risk due to its lack of mitigation infrastructure and strategies.

Due to safety concerns, the students could not enter the river for measurements and focus instead on perceptions of flood risk and evidence of flood mitigation infrastructure. At each site, they collected data from 8 residents or business owners through a questionnaire survey. They also recorded visible flood defenses and land use features at each location through observation.

Resource 1 shows the area around the Klang River where the students conducted their investigation. Resources 2A, 2B, and 2C present the three investigation sites, highlighting the flood defences in place and the observed land use. Resource 3 contains the questionnaire used by the students to gather perceptions of flood risk at all three sites. Resource 4 provides a table summarising the survey results collected from across the three sites.

(a) With reference to Resource 1, explain why the students' hypothesis is appropriate for the investigation. [3]

Award 1 mark for each explanation of why the students' hypothesis is appropriate, to a maximum of 3 marks.
Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Possible responses include:

1) Suitable → The study area (Klang River, Selangor) is a flood-prone region, with settlements located close to the river channel and in close proximity to one another. This makes it easy for students to investigate relative flood risk across different sites. [1]

2) Clearly defined → The three selected sites (Jalan Tepi Sungei, Jalan Tengku Kelana, Kampung Datuk Dagang) have different levels of flood defenses and land uses, which allow students to test the hypothesis. [1]

3) Capable of research → The hypothesis includes measurable variables: the dependent variable (level of flood risk, as indicated by perceptions and impacts) and the independent variable (extent of mitigation infrastructure and strategies in place). [1] This relationship is observable and can be tested using survey and field data. [additional 1 mark]

(b) With reference to Resources 2B and 2C, explain how the students could minimise potential risks when visiting the two sites. [6]

Award 1 mark for each explanation of how the students could minimize potential risks when visiting the two sites, to a maximum of 6 marks.
Award a maximum of 1 additional mark for further development of each explanation, where applicable.
Award a maximum of 3 marks for either site 2B or site 2C.

Possible responses include:

1) Site 2 (Jalan Tengku Kelana, Resource 2B)

- The riverbank walkway and green space may be slippery or uneven. Students should wear proper footwear to prevent falls. [1]
- Students should also remain on designated pedestrian walkways and avoid climbing on or standing close to the retaining walls, as water levels may rise unexpectedly. [1]
- To reduce risk of accidents, students should avoid entering drains or informal flood channels, sticking only to safe, paved walkways to minimize accidents. [1]

2) Site 3 (Kampung Datuk Dagang, Resource 2C)

- Flood-prone, low-lying streets could have standing water or poor drainage after rainfall; students should plan visits in dry weather and monitor forecasts. [1]
- Students should work in pairs/groups to ensure safety from traffic and keep to the sides of the road. [1]
- Students should work in pairs or groups and not alone as the residential area seems to not be too populated and stranger danger might be a risk.[1]
- They should also be on the lookout for wild dogs and avoid provoking them in case they attack [1].

3) General fieldwork safety measures (applies to both sites): They should also carry basic first-aid kits and maintain communication with supervising teachers. These precautions minimise physical risks and ensure rapid response in emergencies. [1] Hydration and sun protection are also necessary as fieldwork may involve extended periods outdoors. [1]

(c) Suggest suitable sampling methods that the students could adopt for their survey at Sites 2A and 2C respectively. [6]

Award 1 mark for each suggestion of suitable sampling methods that the students could adopt for their survey at Sites 2A and 2C respectively, to a maximum of 6 marks.
Award a maximum of 1 additional mark for further development of each suggestion, where applicable.
Award a maximum of 3 marks for either site 2A or site 2C.

Possible responses include:

1) Systematic sampling at Site 2A (Jalan Tepi Sungei): This site looks like a playground located alongside along the river (Resource 2A). Students could approach every third or fifth person along the stretch. [1] This ensures regular coverage of the area, avoids interviewer bias in choosing respondents, and is practical to implement given time constraints. [additional 1 mark]. To ensure that they cover all age groups, the students can also do systematic sampling on a stratified group, meaning the 3rd or 5th person of certain age groups. [additional 1 mark]

2) Stratified random sampling at Site 2C (Kampung Datuk Dagang): This residential estate consists of households located at varying distances from the river (Resource 2C). The sample could first be divided into strata based on proximity to the river (e.g., houses near the river vs. further inland), [1] before randomly selecting respondents within each group. This ensures that the data captures different levels of exposure to flood risk, improving representativeness. [additional 1 mark] Within each group they students can also ensure that a certain number of respondents do the survey so that all the groups have equal representation [additional 1 mark].

(d) With reference to Resource 3, explain the limitations of the survey on perceptions of flood risk in improving the students' investigation. [6]

Award 1 mark for each explanation of the limitations of the survey on perceptions of flood risk in improving the students' investigation, to a maximum of 6 marks.
Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Possible responses include:

1) Reliance on subjective perceptions rather than objective data: Questions such as Q1 ("How serious do you think flood risk is?") and Q5 ("How effective do you think these flood protection measures are?") rely on personal judgement. [1] Residents' answers may be influenced by recent experiences or emotions rather than actual flood frequency or infrastructure effectiveness. This reduces the reliability of findings for assessing real flood risk. [additional 1 mark]

2) Assume respondents know about the various flood mitigation measures listed in Q4. Respondents may not know what the measures are. [1] Instead, students should supplement with necessary photos to help respondents have a better understanding of the various measures. [additional 1 mark]

3) Limited representativeness due to small and uneven sample size: The survey was conducted with only 8 respondents per site. Such a small number may not capture the diversity of experiences and opinions within larger communities, [1] especially in densely populated residential areas like Kampung Datuk Dagang (Resource 2C). As a result, the conclusions drawn may not reflect the wider population's perceptions. [additional 1 mark]

4) Survey design restricts depth of responses: Most questions are closed-ended with fixed categories (e.g., Q4 on mitigation measures), which limit respondents' ability to share unique insights. [1] Although Q6 is open-ended, it is vague and may produce inconsistent answers that are hard to analyse systematically. This constrains the

usefulness of the data for understanding local knowledge and coping strategies. [additional 1 mark]

(e) Describe <u>two</u> other suitable data collection methods that could help improve the students' investigation. [4]

Award 1 mark for each description of a suitable data collection method that could help improve the students' investigation, to a maximum of 4 marks.
Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Possible responses include:

1) Secondary hydrological data: obtain rainfall and river discharge records from local authorities to identify frequency and severity of floods over time. [1] By calculating the recurrence interval, it provides an objective basis for assessing risk. [additional 1 mark]

2) Data from interviews: students can interview residents and find out more about their background and how long they have lived or how often they visit the place to ascertain the validity of their viewpoints [1]. This also gives the students a chance to find out more about the flood risk of that place. [additional 1 mark]

3) Photographic or sketch mapping evidence: Systematically documenting flood defences, drainage, and land use along the riverbank [1]. This would allow comparison across sites and strengthen observations of physical vulnerability. [additional 1 mark]

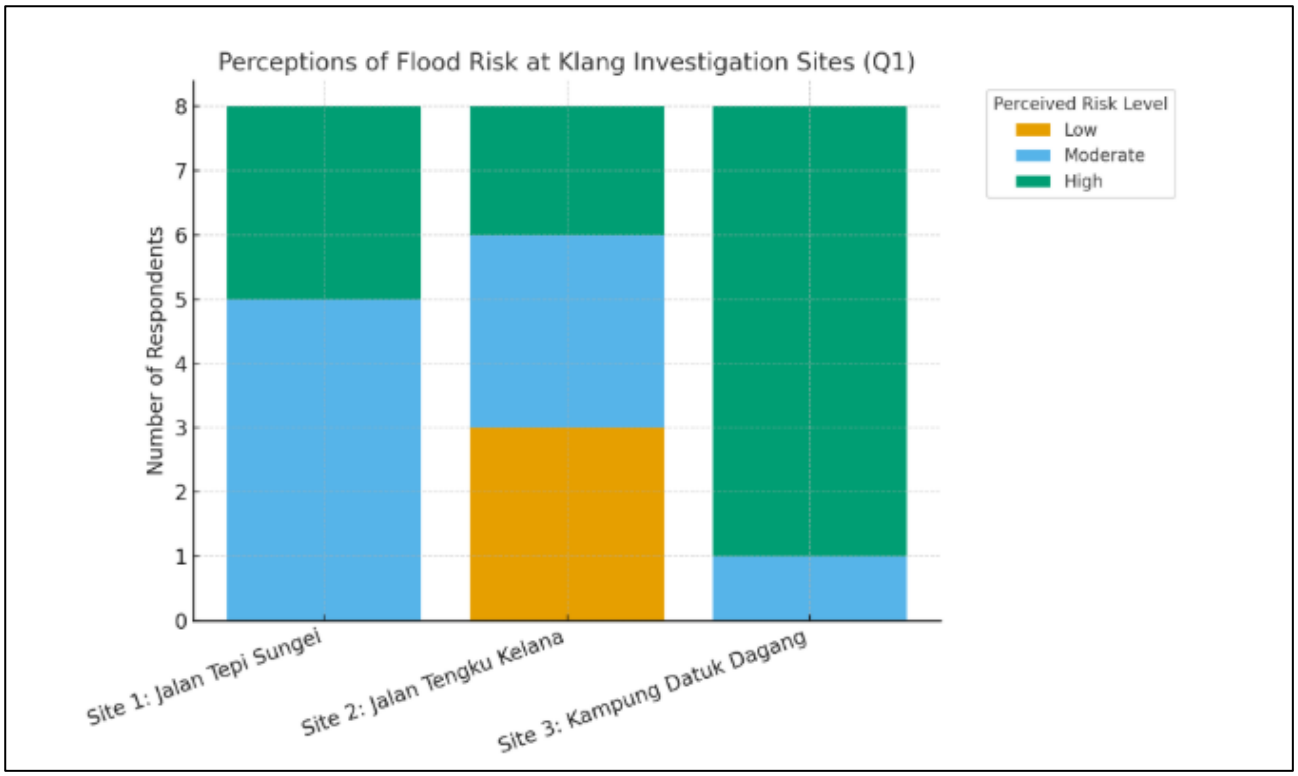
(f) With reference to Resource 4, use an appropriate data representation technique to present the data for Question 1 and explain why it is suitable. [5]

Award a maximum of 3 marks for the appropriate data representation method.
Award a maximum of 2 marks for the explanation of why it is suitable.

Possible responses include:

Suggested method: A stacked bar chart with the three sites on the x-axis and proportions of "Low", "Moderate", and "High" flood risk on each bar. [1+1]

Justification:
- The stacked bar chart allows direct comparison of perceptions of risk across sites. [1]
- It shows that Site 3 (Kampung Datuk Dagang) has overwhelmingly higher perceptions of flood risk (7 "High") compared to Sites 1 and 2, where responses are more evenly distributed. [1]
- It is more effective than a pie chart, as it facilitates cross-site comparison of proportions. [1]



(g) With reference to all the resources, evaluate the validity of the students' conclusion that their hypothesis is true. [10]

Possible Approaches:

Candidates could evaluate the investigation by reviewing the methods used to collect data and the data collected in Resource 4.

For data collection methods, candidates could review the accuracy and reliability of data collected. Evaluation of these methods could include looking at the sample size, the sites chosen for the investigation, etc.

For the data collected in Resource 4, candidates could consider the clarity of the terms used in the questionnaire survey and how these could give rise to differing interpretations by the residents. Candidates could consider how such differing interpretations could affect the conclusions that could be drawn from the analysis of these data.

Candidates could also evaluate their conclusion based on evidence shown in Resources 2A to 2C on the visible flood defenses and land use features at each location.

Candidates could also evaluate other factors that may affect flood risk such as elevation of locations [topography] and availability of insurance coverage etc.

Levels marked using Generic Level Descriptors for H2 fieldwork evaluative questions

Possible responses include:

The students concluded that *Kampung Datuk Dagang faces the highest flood risk due to its lack of mitigation infrastructure and strategies.* This conclusion is largely valid, but not fully conclusive, as the evidence from the resources shows both strong support and important limitations.

Largely valid:

- Perceptions of flood risk (Resource 4, Q1): At Kampung Datuk Dagang, 7 out of 8 respondents rated flood risk as “High”. This is much higher than Jalan Tepi Sungei (3 “High”) and Jalan Tengku Kelana (2 “High”). The dominance of “High” responses supports the claim that Site 3 faces greater risk.

- Past experiences of flooding (Resource 4, Q2–Q3): All 8 respondents at Kampung Datuk Dagang had experienced flooding in the past 5 years, with serious impacts such as property damage (7), loss of income (6), health/safety concerns (4), and displacement (5). This suggests both high frequency and severity of floods at Site 3, unlike Sites 1 and 2 where most respondents had not experienced flooding.

- Lack of mitigation infrastructure (Resource 4, Q4): At Site 3, 5 respondents indicated “None” for mitigation measures, with only 2 noting drainage upgrades and 1 mentioning a warning system. In contrast, Sites 1 and 2 reported stronger defences such as embankments, retaining walls, and drainage upgrades. They could be a possibility why Kampung Datuk Dagang faces the highest flood risk.

- Observational evidence (Resource 2C): The photo of Kampung Datuk Dagang shows a residential estate of landed houses with no visible structural flood defences. By comparison, Resource 2B (Jalan Tengku Kelana) shows retaining walls and green buffers, while Resource 2A (Jalan Tepi Sungei) indicates zoning along the riverbank. This visually confirms weaker protection at Site 3.

Not fully valid

- Perceptions may not match actual hydrology (Resource 3): Survey questions (e.g., Q1 and Q5) rely on subjective perceptions, which may exaggerate risk based on recent flooding events. Without objective river discharge or rainfall data, conclusions about absolute flood risk remain tentative.

- Small sample size (Resource 4): With only 8 respondents per site, the findings may not be representative of the wider community. The apparent dominance of “High” responses at Site 3 could reflect sampling bias rather than the full population’s experience. Improvement: they should sample more people to have a better representation.

- Risk present at other sites too: Resource 4 shows that even at Site 1, 3 respondents rated flood risk as “High”, and 2 households experienced flooding. Similarly, at Site 2, property damage (2 respondents) and safety concerns (2 rated as “high”) were reported despite the presence of retaining walls. This indicates that flood risk is not exclusive to Kampung Datuk Dagang.

- Flood risk is affected by a multitude of factors and students should collect other data to supplement findings such as elevation of locations [topography] and availability of insurance coverage etc.

Overall evaluation:

The conclusion that Kampung Datuk Dagang faces the highest flood risk due to weaker mitigation is valid, supported by strong survey evidence (Resource 4) and visual observation (Resource 2C). However, it is not fully conclusive. The reliance on small-sample perception surveys (Resource 3) mean that the conclusion may be oversimplified. Resource 1 also shows all three sites are located directly by the Klang River, so riverine flooding is a shared hazard, not unique to Kampung Datuk Dagang.

Section B

Cluster 3 Sustainable Future and Climate Change

2 Resource 5 shows the proportion of Singapore's aged population (above 65) in 2010 and 2018. Resource 6 is an infographic on ageing in Seoul and Singapore. Resource 7 is a photograph showing the facilities in Shinnae Medical Housing in Seoul.

(a) Using Resource 5, describe the changes in the spatial distribution of the aged population (above 65) in Singapore between 2010 and 2018. [4]

Award 1 mark for each description of the changes in the spatial distribution of the aged population (above 65) in Singapore between 2010 and 2018, to a maximum of 4 marks. Award a maximum of 1 additional mark for further development of each description, where applicable.

Possible responses include:

- Between 2010 and 2018, Singapore experienced a significant increase and a wider distribution of its aged population (above 65). [1]

- From the oldest towns like Bedok and Toa Payoh to other mature estates like Bishan, Tampines and Serangoon, Singapore saw a significant intensification of its aged population between 2010 and 2018. [1]

- East Region: Tampines and Pasir Ris, which had moderate numbers in 2010, experienced substantial growth by 2018, with many zones moving into the higher 4,000 - 8,000 resident categories.

- North Regions: Woodlands and Yishun also experienced a notable increase, with more of their sub-zones moving into higher population brackets (4,000 - 8,000) for seniors.

- New Areas like Punggol and Sengkang, which had few elderly in 2010 (mostly under 200), saw an increase in elderly population by 2018, showing spatial spread into northeastern estates. [1]

- West Region: Jurong West, Jurong East, Bukit Batok and Choa Chu Kang showed a similar pattern of growth, with many areas advancing from the 200 - 2,000 range to the 4,000 - 8,000 range.

(b) Explain the changes in the spatial distribution of the aged population (above 65) in Singapore between 2010 and 2018 shown in Resource 5. [5]

Award 1 mark for each explanation of the changes in the spatial distribution of the aged population (above 65) in Singapore between 2010 and 2018, to a maximum of 5 marks. Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Possible responses include:

- Significant Overall Growth - There was a notable increase in the aged population across Singapore reflecting Singapore's national ageing trend. [1] This is largely due to longer life expectancy and the post-war baby boomers entering retirement, causing the elderly share of the total population to expand rapidly. [1 additional mark]

- Ageing-in-Place in Mature Estates - The deepening concentration of seniors in older towns is a direct result of residents ageing-in-place over the years. [1] The early homeowners who moved into estates like Bishan, Tampines and Serangoon have now reached their senior years, choosing to remain in familiar communities with strong social ties and established amenities. [1 additional mark]

- Emergence of New Ageing Zones - The significant growth of the elderly population in other established towns in the north, east and west regions reflects a subsequent wave of the ageing-in-place trend. [1] The shift shows how the first generation of residents in these estates has started to age, creating new ageing "pockets" beyond the traditional mature estates. [1 additional mark]

- More Widespread Distribution: The appearance of elderly residents in new towns like Punggol and Sengkang is largely due to a desire for inter-generational family support and housing monetization schemes. [1] Government policies like the Proximity Housing Grant (PHG) incentivize seniors to move to be near their adult children in newer estates like Punggol, while some also "right-size" from larger, older flats to unlock their cash value. [1 additional mark]

(c) Using Resource 6, compare the ageing trends in Seoul and Singapore. [6]

Award 1 mark for each comparison of the ageing trends in Seoul and Singapore, to a maximum of 6 marks. Award a maximum of 1 additional mark for further development of each comparison, where applicable.

Possible responses include:

Similarities

- Both Seoul and Singapore are experiencing similar ageing trends, including a declining old-age dependency ratio, a rise in seniors (especially females) living alone, and an increasingly educated senior population. [1]

- Declining Old-Age Dependency Ratio [1]
  - Both cities are experiencing a drop in their old-age dependency ratio. In Seoul, the ratio declines from 5.6 today to 2.8 by 2030, and in Singapore, from 5.1 to 2.1. [1 additional mark]

- More Educated and Higher-Income Seniors [1]
  - Both cities show that baby boomers today are more educated. In Seoul, 45.5% of seniors attended technical college or higher, while in Singapore, 58% completed at least secondary school. [1 additional mark]
- Rise of Seniors Living Alone, Especially Females [1]
  - In both cities, a growing number of seniors live alone, with women forming the majority. In Seoul, 27.7% of seniors live alone (68.5% female), while in Singapore, the figure is 8.9% (66% female). [1 additional mark]

Differences

- A key difference lies in the specific nature of their distinct aging challenges, with Seoul facing socio-economic issues while Singapore confronts healthcare-related problems. [1] For instance, Seoul's distinct trends include early retirement and intergenerational conflict for jobs, whereas Singapore's trends are an increasing chronic disease burden and issues of caregiving. [1 additional mark]

- Furthermore, the prevalence of seniors living alone is significantly higher in Seoul compared to Singapore. [1] In 2015, 27.7% of seniors in Seoul lived alone, which is more than three times the 8.9% reported for Singapore in the same year. [1 additional mark]

- Lastly, the educational profile of seniors is measured differently, indicating a potential variance in the level of educational attainment between the two cities.[1] The data shows 45.5% of Seoul's baby boomers attended technical college or higher, while 58% of Singapore's baby boomers have at least secondary school education. [1 additional mark]

(d) Explain the challenges faced by the elderly in Seoul as shown in Resource 6. [6]

Award 1 mark for an explanation of the challenges faced by the elderly in Seoul, to a maximum of 6 marks. Award 1 additional mark for further development of the challenges faced by the elderly in Seoul, where applicable.

Possible responses include:

- High Rate of Seniors Living Alone:
  - With 27.7% of seniors living alone, especially elderly women (68.5%), social isolation becomes a major concern. [1] For these elderly women, the risk of loneliness is magnified by the financial and employment struggles, placing them at a higher risk of poverty, neglect, and severe mental and physical health issues. [1 additional mark]

- Early Retirement and Financial Insecurity:

- Due to a significant gap between retirement age (52.6 years) and pension eligibility (65 years), many elderly in Seoul face financial instability. [1]. This forces a significant portion of the elderly population into a precarious 12-year "income gap" where they are no longer formally employed but are not yet eligible for their state pension, making them highly vulnerable to poverty. [1 additional mark] OR This trend highlights the lack of income security in later life especially if they are single or do not have children to support them in their retirement years. [1 additional mark]

- Intergenerational Conflict:
  - Compounding this financial insecurity is intense intergenerational conflict in the job market. Seniors trying to bridge the income gap are forced to compete directly with younger people for available jobs. [1] This not only creates social tension but also puts older workers at a significant disadvantage, making it extremely difficult for them to secure stable employment after being pushed into early retirement. [1 additional mark]

(e) Using Resource 7, explain how the different facilities help to meet the social and psychological needs of the elderly residents. [5]

Award 1 mark for each explanation of how the different facilities help to meet the social and psychological needs of the elderly residents, to a maximum of 5 marks. Award a maximum of 1 additional mark for further development of each explanation, where applicable.
Award a maximum of 3 marks for explanation of either social or psychological needs.

Possible responses include:

Top left photo
- The facility helps meet the social needs of its elderly residents by organizing group activities that prevent loneliness and foster a sense of community. [1] The image shows residents gathered together in a common area for health check-ups, providing a valuable opportunity for them to interact with peers and staff rather than remaining isolated in their rooms. [1 additional mark]

- The facility also addresses the residents' psychological needs by providing tangible proof that their health and well-being are being actively cared for. [1] By conducting regular health monitoring, such as the blood pressure check depicted, the facility offers residents a profound sense of security and reassurance, which can significantly reduce anxiety about their health. [1 additional mark]

Top right photo
- This facility helps meet the social needs of the elderly by providing a dedicated common area that encourages spontaneous interaction and shared activities. [1] The community zone, equipped with a central dining table and a television, serves as a natural gathering point for residents to share meals, watch programs, and engage in conversation, reducing feelings of isolation. [1 additional mark]

- This space also supports the psychological needs of residents by creating a comfortable, home-like environment that fosters a sense of normalcy and belonging. [1] The presence of familiar domestic elements like a refrigerator, a dining set, and

slippers neatly arranged at the entrance helps to reduce the institutional feel of the facility, promoting comfort and mental well-being. [1 additional mark]

Bottom left photo

- This facility supports the psychological needs of residents by empowering them with a sense of autonomy and control over their personal health. [1] By providing accessible equipment for self-monitoring, as shown in the image, residents can proactively track their own health data, which fosters independence and can reduce the anxiety associated with dependency. [1 additional mark]

- The health station also helps meet social needs by creating a functional space for informal interaction and mutual support. [1] The image shows two people present at the monitoring equipment, illustrating how these routine checks can become opportunities for residents to connect with staff or peers, share encouragement, and engage in casual conversation. [1 additional mark]

Bottom right photo

- The rooftop garden helps meet the social needs of residents by creating a natural and relaxed setting for community building. [1] As residents tend to their individual garden lots side-by-side in the communal "Sky Garden," they have frequent opportunities to interact, share gardening tips, and form friendships over a shared interest. [1 additional mark]

- This facility also meets crucial psychological needs by providing residents with a sense of purpose, autonomy, and accomplishment. [1] The caption states that "Each resident is assigned a lot," which gives them personal responsibility and control, allowing them to feel pride and satisfaction from nurturing plants and seeing them flourish. [1 additional mark]

(f) With reference to Resource 7, propose ways in which Shinnae Medical Housing in Seoul can help meet the economic needs of its elderly residents.[4]

Award 1 mark for each proposal in which Shinnae Medical Housing in Seoul can help meet the economic needs of its elderly residents, to a maximum of 4 marks.
Award a maximum of 1 additional mark for further development of each proposal, where applicable.

Possible responses include:

- The facility can create structured, flexible part-time job opportunities on-site for its residents. [1] This could involve hiring residents for paid roles such as assisting with gardening, helping with administrative tasks, or coordinating activities, providing them with a regular stipend and a dignified way to supplement their income. [1 additional mark]

- The facility should host regular financial literacy workshops tailored to the specific economic situations of seniors. [1] These sessions would empower residents by teaching them how to manage a fixed income, navigate complex pension or subsidy schemes, and identify and avoid financial scams that often target the elderly. [1 additional mark]

- A cooperative can be established to leverage the lifelong skills of residents for economic benefit. [1] Through this program, residents could teach workshops in areas like cooking or crafts to the wider community for a fee or offer their services to fellow residents, creating both income-generating and cost-saving opportunities. [1 additional mark]

- The facility can develop the rooftop Sky Garden into a small-scale social enterprise that generates income for residents. [1] This initiative would involve residents cultivating produce for sale at a local market or creating value-added **products like** kimchi and jams in the Community Zone, with the profits being shared among the participants. [1 additional mark]

Section C

Answer EITHER question 3 OR question 4.

3 'Community participation is the key factor in effective slum management.' Evaluate this statement. [20]

Candidates could also approach the question by making a judgement on the relative importance of community participation compared to other factors influencing slum management outcomes, such as the political will of local authorities, the scale of the issues faced, the level of financial and technical resources available, and the role of external stakeholders such as NGOs and private developers. They could examine how participation interacts with these factors, for example, whether well-funded projects without genuine community input can still succeed, or whether active participation can overcome resource limitations. In their evaluation, candidates can consider whether participation is necessary but not sufficient on its own, and whether its effectiveness depends on targeted, inclusive, and context-sensitive approaches that ensure the diverse needs of stakeholders are met.

Candidates could approach the question by making a judgement on whether community participation is a key factor in successful slum management, through an evaluation of two or more case studies. They could assess the ways in which participation — such as the inclusion of residents in decision-making, design, and implementation — helps address the complexity of slum issues, including high population density, insecure land tenure, informal governance, poor infrastructure, and socio-economic marginalisation. Candidates could also evaluate examples where participation was limited or tokenistic, and how this constrained the effectiveness of management strategies. In their analysis, they could compare contexts where strong community involvement improved outcomes, such as greater ownership of housing upgrades, improved tenure security, and enhanced access to services, with those where exclusion from the process led to resistance, maladaptation, or the failure to maintain facilities over time.

4 Evaluate the influence of natural factors on the Earth's climate since 1850. [20]

Candidates could approach the question by making a judgement on whether natural factors such as fluctuations in solar irradiance, movement of the thermohaline and

short-term oscillations like El Niño and volcanic eruptions have been the main drivers of climate change since 1850, or whether anthropogenic factors dominate. Candidates may analyse the timing of volcanic eruptions (e.g., Krakatoa 1883, Mount Pinatubo 1991) that produced short-term cooling due to aerosols blocking incoming solar radiation, and consider how these effects compare with the long-term warming trend. Candidates could also examine variations in solar output, including sunspot cycles, and evaluate whether these fluctuations correspond closely with periods of warming or cooling. Candidates may weigh these against evidence from the instrumental record since the 1850 to see how far they correspond with variations in natural factors or changes in human activities over a similar period.

Candidates could also approach the question by making a judgement on whether natural factors play a more significant role in driving or moderating changes in Earth's climate since 1850. Candidates may consider how natural factors (such as volcanic eruptions, variations in solar output, or ocean–atmosphere oscillations like El Niño) have contributed to short-term climatic fluctuations, and weigh this against the extent to which these same factors may have acted to offset or moderate long-term warming trends. Candidates could analyse this influence by considering spatial scale, where natural events may cause either global changes (e.g. volcanic eruptions reducing worldwide temperatures) or more localised impacts (such as El Niño bringing drought to Southeast Asia). Candidates could also examine temporal scale, since most natural influences are short-lived, lasting only a few years, in contrast to the long-term and sustained warming linked to human activity. Candidates could also evaluate whether natural factors on their own are sufficient to explain the continued rise in global temperatures since the mid-20th century, or whether anthropogenic factors provide a more convincing explanation.